AAEE MINIMONOGRAPH #28

Nerve injury triggers a sequence of electrophysiological events dependent upon changes occurring both at the neuromuscular junction and the muscle membrane. The time course of these physiologic and morphologic alterations is reviewed in light of recent information provided by studies of extrajunctional acetylcholine receptors, axoplasmic transport, and freeze-fracture electron microscopy. A thorough knowledge of these changes is essential to understanding the clinical impact of nerve injury.

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AAEE MINIMONOGRAPH #28: INJURY TO PERIPHERAL MOTOR NERVES

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Injury to a peripheral motor nerve produces a predictable pattern of electrophysiological events. The nature of the injury is a critical factor in determining the severity of the injury and the prognosis for full recovery. Compressive neuropathy, whether acute or chronic, is predominantly demyelinating in many cases and is often associated with rapid and relatively complete recovery.[45] On the other hand, nerve transection, stretch, or crush is followed by Wallerian degeneration, more delayed and incomplete recovery, and distinctive electrophysiological responses in the distal motor unit in the early period after injury. A thorough knowledge of the orderly sequence of the events accompanying Wallerian degeneration and regeneration permits the electromyographer to make clear statements about diagnosis and prognosis. In this review we will consider the histological and electrophysiological changes following a nerve injury that produces mainly axonal degeneration, with particular emphasis upon the alterations at the neuromuscular junction and in the muscle membrane.

Distal to an axonal transection, the light microscopic changes of Wallerian degeneration (first described by Waller in 1850) are similar in all species and have recently been reviewed.[55,61] The axoplasm undergoes disruption, and there is loss of longitudinal orientation with fragmentation of neural tubules and neurofilaments throughout the axon. Mitochondria become swollen and discontinuities develop in the axolemma. Areas of swelling and narrowing occur in the axon, and fragmentation takes place initially between nodes of Ranvier, usually in the smallest fibers first. There is some controversy about whether changes in the axon progress simultaneously along the entire distal segment or whether they move centrifugally from the site of nerve injury. Degenerating axonal fragments are surrounded by myelin, and ellipsoids are formed that subdivide into smaller spheres. Proliferating Schwann cells and macrophages produce further degradation by approximately the fifth day. Digestion of these fragments occurs over the next several weeks. The same sequence of events occurs whether the nerve is injured by a crush, stretch, or complete section.

EFFECTS OF NERVE INJURY ON THE NEUROMUSCULAR JUNCTION

Before these changes of Wallerian degeneration in a damaged axon become apparent, abnormalities at the neuromuscular junction are detectable.[27,46] Following nerve transection in primates, there is a progressive fall in amplitude of both muscle and nerve action potentials on successive days. Transmission failure occurs first at the neuromuscular junction. The muscle response to nerve stimulation disappears by the fourth or fifth day after nerve section, whereas the ascending nerve action poten-

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tial is obtainable for 5–8 days.[26] In rabbit, guinea pig, cat, dog, baboon, and human, neuromuscular transmission fails before axonal conduction fails.[29] In general the time to neuromuscular transmission failure is longer in humans compared with lower mammals.[21,28,33,40,52]

Studies of the human facial nerve after transection have documented neuromuscular transmission failure beginning on the fifth day and complete by the eighth day (Fig. 1).[28] In more recent studies of human motor nerves undergoing Wallerian degeneration in the limbs, neuromuscular transmission failure was detectable on the fourth day and complete by the seventh day.[66] Changes in sensory nerves were delayed by 2 or 3 days but then followed a similar time course.[66,67]

Based on these few observations, it would appear that the length of the distal stump may not be a major factor in defining the onset of neuromuscular transmission failure. However, this relationship in man deserves further study, since animal studies have documented a definite link between distal stump length and neuromuscular junction changes as described below.

Miledi and Slater[44] correlated the functional and structural changes at the neuromuscular junction following nerve section in the rat. Neuromuscular transmission was normal until 8–10 hours after nerve section, at which point an increasing number of endplates began to fail, and after 20 hours no endplates transmitted an impulse. Transmission failure occurred abruptly at most endplates and was usually accompanied by cessation of spontaneous miniature endplate potentials, although occasionally miniature endplate potentials persisted after junctional transmission had failed. There was a direct relationship between the time of junctional failure and the length of the degenerating nerve stump. For each additional centimeter of nerve in the distal stump, neuromuscular transmission failure was delayed 45 minutes. Ultrastructural changes in nerve endings closely paralleled changes in function. There were no detectable morphological abnormalities at the endplate in the first 8–12 hours after nerve section. When neuromuscular transmission was failing, some endplates were clearly undergoing structural breakdown. All endplates were morphologically abnormal when neuromuscular transmission failure was complete.

The development of freeze-fracture electron microscopy has made possible the study of large areas of the endplate and structures within the nerve terminal membrane that had otherwise been inaccessible. The active zones on the nerve terminal

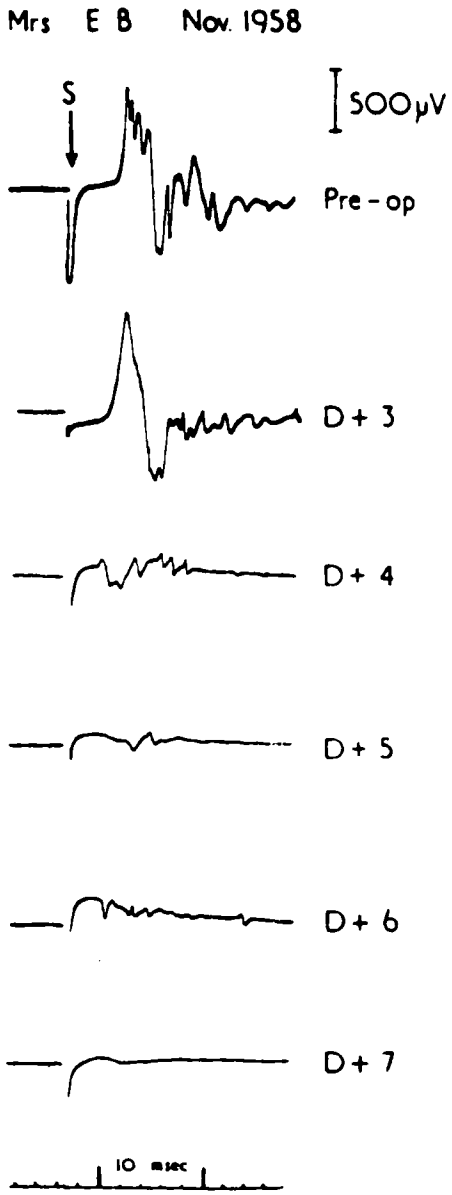


FIGURE 1. Tracings showing muscle response to nerve stimulation before and on successive days after nerve section. The stimulus (S) is shown by artifact in each trace and is followed by muscle action potentials up to the sixth postoperative day (D+6) but not after this time. After Gilliatt.[28]

are the sites for the quantal release of neurotransmitter by synaptic vesicle exocytosis at the neuromuscular junction.[34] The active zones lie directly opposite the junctional folds of the postsynaptic membrane and are usually perpendicular to the longitudinal axis of the nerve terminal. After nerve section, and before neuromuscular transmission failure, no histological abnormality is evident in the active zones or in the remaining portion of the neuromuscular junction with freeze-fracture elec-

tron microscopy.[37] As neuromuscular transmission begins to fail, the active zones become disrupted. The particles that normally occur in two double rows become disorganized and interrupted and finally vanish. Eventually, Schwann cells completely engulf nerve terminals, so that the junctions have a new cellular arrangement (Schwann cell and muscle fiber), and neither endplate potentials nor miniature endplate potentials can be recorded from the muscle fiber. Ridges in the Schwann cell occur directly opposite the junctional folds and perpendicular to the longitudinal axis of the junctions, but no organized rows or large particles (corresponding to the active zones at nerve terminals) are found along the Schwann cell ridges.

The disruption of nerve terminal active zones is probably associated with the failure of endplate potentials, and the disappearance of miniature endplate potentials probably corresponds to the engulfment of nerve terminals by Schwann cells. The loss of neuromuscular transmission in the degenerating neuromuscular junction appears to be entirely due to changes at the nerve terminal, since the morphology of the junctional folds and the endplate sensitivity to acetylcholine remain unchanged when transmission first fails after denervation.[4]

GENERAL EFFECTS OF DENERVATION ON MUSCLE

Denervation is associated with considerable muscle fiber atrophy, and the myofibrillar atrophy is proportional to the reduction in muscle fiber size.[20] During the first few days after denervation, there is an increase in mitochondrial volume and sarcotubular surface, whereas fiber size and contractile elements decrease rapidly. Within 1 week of denervation, mitochondria and sarcotubules decrease in amount, but less than the decrease in contractile elements, and the overall effect is that of an increase in the concentration of sarcotubular profiles. Mitochondrial volume returns toward normal during the first month after denervation. The rate of muscle fiber atrophy after denervation varies in different muscles and with different species. In rat limb muscles there may be a 50% reduction in weight within 2 weeks of denervation,[58] whereas rat diaphragm may transiently hypertrophy before atrophy occurs.[60] The rapid increase in muscle DNA content after denervation may be attributed to mitosis in interstitial cells, but labelled DNA precursor molecules appear also in satellite cells, and mononucleate myoblasts are seen in interstitial spaces. The general effect of denervation is to decrease the rate of synthesis of muscle proteins and to increase their rate of degradation.[30]

OTHER EFFECTS OF DENERVATION ON SKELETAL MUSCLES

In addition to altered neuromuscular transmission and obvious muscle atrophy, there are at least five other changes in muscle membrane properties that follow denervation. The earliest muscle change that occurs after denervation is a reduction in the resting membrane potential. Soon afterwards, tetrodotoxin fails to block muscle action potentials initiated near old motor endplates, and, within a week after denervation, action potentials can be evoked in any part of a muscle fiber despite the presence of tetrodotoxin.[51] Next there is an increase in the sensitivity of extrajunctional regions of the muscle membrane to acetylcholine. Spontaneous fibrillation activity arises from the endplate region, with a latent interval that depends upon length of the distal nerve stump. Finally, an increase in the specific membrane resistance of the muscle develops. Three of these events will be considered in some detail: (1) decrease in resting membrane potential, (2) fibrillation potentials, and (3) increased sensitivity to acetylcholine.

Decreased Resting Membrane Potential. This subject has been reviewed recently.[63] In denervated mouse and rat skeletal muscles, the reduction in resting membrane potential (which occurs after cessation of miniature endplate potentials) amounts to approximately 15–20 mV. Using the soleus muscle of the intact rat, highly reproducible results have been obtained with regard to alterations in the resting membrane potential after denervation.[59] The resting potential begins to decrease 18–24 hours after nerve transection and reaches a nadir after about 42 hours, with no further significant change for the next 30 hours in experiments performed with a nerve stump of less than 2 mm. Whereas with the short nerve stump miniature endplate potential activity had virtually ceased at 14 hours and decrease in the muscle resting potential occurred at 18–20 hours, with a longer nerve stump of 40 mm the resting potential did not begin to decrease until 22 hours. Because the denervated muscles are inactive regardless of the length of the nerve stump, some factor other than contractile activity must account for the increasing delay, with increasing stump length, from the time of nerve transection to the decrease in the resting membrane potential. In two different studies the delay was approximately 45 minutes per centimeter of nerve stump, a value

quite close to the rate of rapid axoplasmic transport.[44,59] Since it is likely that axonal transport is necessary for the maintenance of nerve terminals, it is not surprising that changes in nerve terminal function, including neuromuscular transmission and the neural influence on muscle resting membrane potential, should follow a time course corresponding to that of rapid axonal transport.

The changes in resting membrane potential occur first in the endplate region and only later appear in the more distal regions of the muscle.[8] The observed depolarization is caused by a reduction in the electrogenic activity of the sodium pumping mechanism of the muscle membrane, with decreased resting potassium permeability and increased sodium permeability.[63] The myofiber atrophy appearing after denervation reduces the fiber diameter, contributing to an increase in the input impedance of the fiber. The reduced amount of synaptic current that is needed to depolarize the denervated muscle fiber contributes to the increased sensitivity of the muscle cell to both electrical and chemical stimuli. However, both of these changes (the decrease in resting membrane potential and the increased input impedance of the muscle fiber) are minor in comparison with the appearance of acetylcholine receptors outside the synaptic region.[11]

Fibrillation Waves. In clinical work the appearance of fibrillations in human muscles characteristically occurs approximately 2–3 weeks after a nerve injury that results in a functional separation between the nerve terminal and the postsynaptic membrane. Fibrillations may occur as early as 2 days after experimental denervation of rat muscle.[53] Fibrillations represent the spontaneous and usually asynchronous depolarization of individual muscle fibers. Fibrillation waves seen in the clinical electromyography laboratory in patients with nerve injuries indicate the presence of axonal degeneration. The intensity of fibrillation wave activity correlates in an approximate way with the amount of axonal degeneration. Fibrillation wave activity is reduced in cool muscles and completely abolished by ischemia.[25] Increased fibrillation wave activity is observed when extracellular calcium concentrations are reduced.[57] Drugs that block neuromuscular transmission (e.g., curare) have no effect upon fibrillation wave activity, an observation that is consistent with the origin of fibrillations within the muscle membrane.[7] Fibrillation potentials usually arise near old endplates[57] and propagate along the whole length of the muscle fiber.[17] Fibrillations may fire regularly (5–20 Hz) or irregularly (0.1–25 Hz), with the two types occurring equally commonly in denervated muscle.[10] These different discharge patterns appear to reflect different pathophysiologic mechanisms. Regularly discharging fibrillation potentials are triggered by spontaneous oscillations in the membrane potential (Fig. 2). Although the cause of the oscillations is not known, the lowered resting membrane potential that follows denervation appears to be a critically important factor.[62] After the initial discharge, there is progressive lowering of the firing threshold as a result of reduced sodium inactivation, which may reflect the appearance of a new class of sodium channels in the muscle membrane of denervated muscle, a proposal for which there is some evidence.[62] These oscillations and the initiation of regular fibrillations are primarily localized in the region of the old endplate zone, where sodium channel density is greatest.[10]

Irregularly discharging fibrillations result from random, discrete, spontaneous depolarizations of nearly constant amplitude (Fig. 3).[10] The amplitude and frequency of these depolarizations are altered by small changes in membrane potential; and the depolarizations are abolished by lowered sodium and by tetrodotoxin. These subthreshold discharges are presumably due to increased sodium conductance in the membranes of the transverse tubules.[50] It has been shown in vitro that only a proportion of denervated fibers are fibrillating at any given time. Results from organ culture suggest

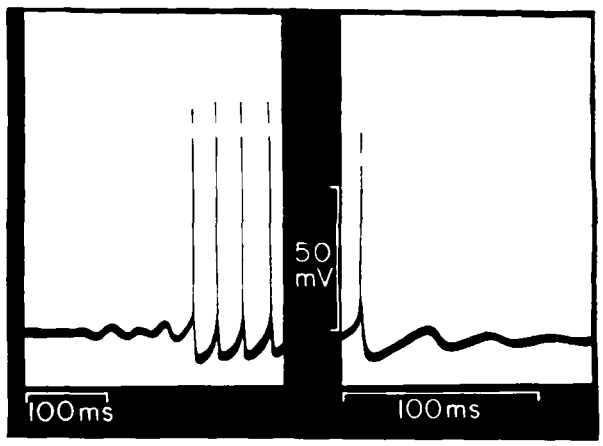


FIGURE 2. Spontaneous fibrillation potentials of the regular rhythmic type recorded in single fibers of the chronically denervated extensor digitorum longus muscle of the rat in vivo. The records show the oscillations that start a train of fibrillation potentials and the damped oscillations following the end of such a train, respectively. The break in the action potential trace indicates the zero potential level of the cell. After Thesleff and Ward.[62]

that the fibrillatory activity is both cyclical and self-inhibiting, with inactive periods that are longer than active periods in a given denervated muscle fiber.[50] The observation that fibrillations are absent in some denervated muscle fibers may result from this rhythmic cycling of activity.[10]

The relationship between fibrillations and denervation supersensitivity is still not completely clear. The time course of the appearance of both phenomena after denervation is identical, yet several important differences between these phenomena remain unexplained. The self-inhibiting action of fibrillations, followed by long periods of inactivity, has a negligible effect on extrajunctional acetylcholine sensitivity.[50] Furthermore, the sodium conductance change underlying fibrillations is blocked by tetrodotoxin and not curare (in doses that block acetylcholine receptors) and is affected by small changes in membrane potential.[50] In order to better understand these differences and the factors responsible for denervation supersensitivity, it is necessary to review our current understanding of extrajunctional acetylcholine receptors.

Increased Sensitivity to Acetylcholine.[24,48] The distribution of acetylcholine receptors in mammalian skeletal muscle is regulated by motor innervation. In innervated muscles acetylcholine (ACh) receptors are almost exclusively located at neuromuscular junctions, but following denervation there is a proliferation of "extrajunctional receptors" at sites outside the endplate.[11] The number of acetylcholine receptor molecules per muscle fiber increases 5- to 50-fold after denervation, with the vast majority of new receptors found in extrajunctional sites.

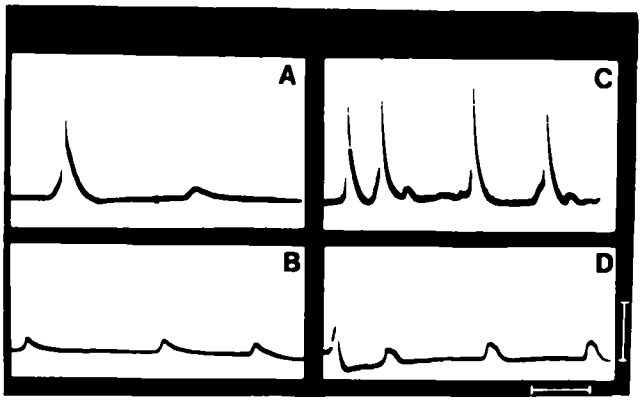


FIGURE 3. Spontaneous, irregular fibrillation potentials triggered by discrete and randomly occurring depolarizations. Records **A** and **B** are from the same fiber, records **C** and **D** are from different fibers in the chronically denervated rat diaphragm muscle in vivo. Time scale for records **A**, **B**, and **D** is 40 msec, time scale for record **D** is 100 msec. Voltage scale is 20 mV. After Thesleff.[62]

The stability following denervation of postsynaptic structures, the persistence of clustered junctional acetylcholine receptors, and the low turnover of these receptors provide evidence against the possibility that junctional acetylcholine receptors redistribute after denervation.[24] The metabolic turnover of junctional acetylcholine receptors is very slow in both innervated and newly denervated muscles. Postsynaptic structures survive for long periods after denervation. Whereas nerve terminals degenerate within 1 week, frog muscle junctional folds and postsynaptic densities remain unaltered for up to 130 days after denervation.[4] Postsynaptic structures in mouse muscles and rat diaphragm survive up to 4 months after denervation. The former postsynaptic areas also retain high acetylcholine receptor density in denervated muscle.[49]

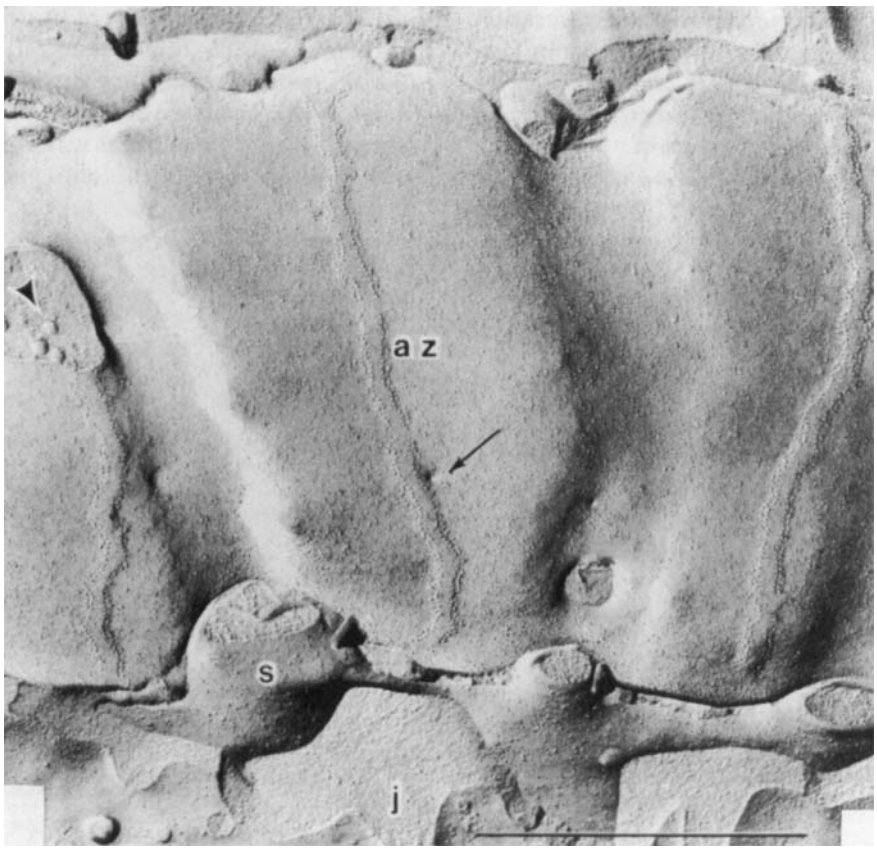
Much more is known about the mechanisms and rates of turnover of extrajunctional receptors compared with junctional receptors. Estimates of average receptor lifetimes for extrajunctional receptors range from 6 to 35 hours, with some of the differences reflecting species or cell line differences. The degradation rate of acetylcholine receptors is sensitive to pH, temperature, and protein environment. Because of the relatively short average lifetime, the entire population of extrajunctional receptors must be replaced every few days.[12] It is assumed that the mechanisms for turnover of junctional and extrajunctional receptors are similar, even though the rate of junctional receptor turnover is much slower.

REPAIR PROCESSES IN PERIPHERAL NERVE

After denervation, reinnervation occurs by (1) sprouting of undamaged axons in close physical proximity to the denervated muscle and (2) by axonal regeneration from the site of nerve injury. In partial denervation sprouting accomplishes reinnervation within a few days, whereas axonal regeneration requires weeks to months, depending upon the type of nerve injury and the distance between the site of nerve injury and the muscle. Regenerating axon nerve terminals displace endplates formed by sprouts from neighboring neurons to complete the process of reinnervation.

Sprouting of Motor Nerves.[5,13,43] Exner in 1884[22] made one of the first references to motor nerve sprouting. He showed that individual muscle fibers were not permanently denervated after partial transec-

tion of the nerve, and he proposed that recovery of muscle function was due to intramuscular (collateral) growth by the remaining intact nerves. It was later demonstrated that partial denervation was followed after several weeks by complete recovery of muscle tension in the absence of reinnervation by severed axons. Edds[18] and Hoffman[35] identified fine nerve outgrowths from the nodes of Ranvier of remaining intact axons (nodal sprouts) in partially denervated muscle. Using a silver stain, Hoffman[35] also demonstrated fine outgrowths from intact nerve terminals (terminal sprouts). These observations confirmed that sprouting was actually new growth from axons that already had functional contact with other muscle fibers, and that activation of previously nonfunctional nerve branches was not involved. *Terminal* sprouts first appear as fine, irregular extensions of nerve terminal branches, best visualized with zinc iodide-osmium tetroxide. These sprouts can arise from a single nerve terminal as well as from the unmyelinated preterminal axonal segment.[1] *Nodal* sprouts arise from nodes of Ranvier within the intramuscular nerves. Not all of the nodes within the nerve of a partially denervated muscle produce visible sprouts.[35] Within 4 days after partial denervation, most final nodal sprouts arise from nodes within 200 μm of the endplates they were innervating.[35] There appears to be a definite clustering of nodal sprouts near points from which denervated nerve sheaths were branched. Long lengths of intramuscular nerve without branch points contain no sprouts, even though they are in close proximity to denervated endplates. The shortest denervated perineural sheaths are the first to become innervated. The major determinant of nodal involvement in sprouting is apparently the distance of the node from a denervated endplate via a vacant perineural sheath.[56] Each nodal sprout finds a vacant perineural sheath leading to the nearest denervated endplate, although the guidance mech-



A

FIGURE 4. (**A**) Nerve terminals (P face) in a normal neuromuscular junction. Active zones (az), with two double rows of large particles, are located opposite junctional folds (j). A portion of this nerve terminal was cross-fractured and revealed synaptic vesicles (arrowhead). A dimple (arrow) is seen next to an active zone. Schwann cell processes (s) separate neighboring active zones. After Ko.[38] (**B**) Nerve terminal (P face) during the initial stage of active zone regeneration, 13 days after denervation. Clusters of large intramembrane particles (arrows) appear aligned with junctional folds (j). In this muscle, 30% of the junctions showed EPPs. Bar = 1 μm. Magnification $\times$45,000. After Ko.[38]

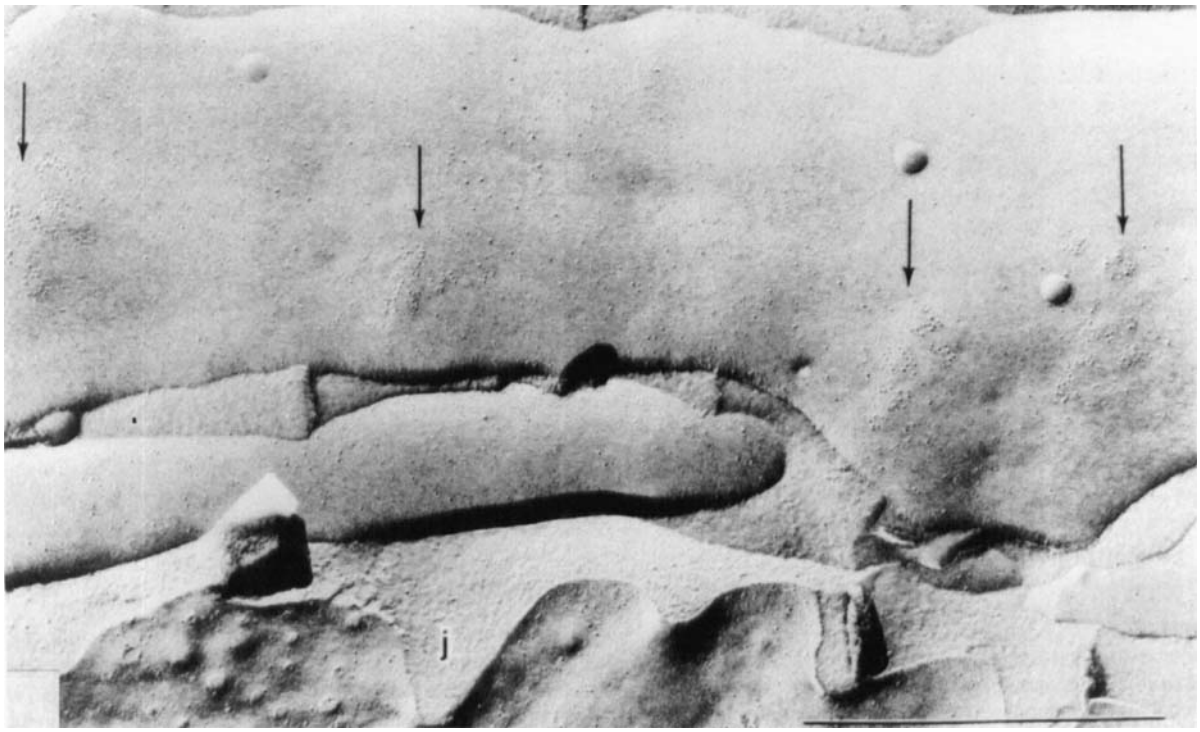
anism is unknown. Terminal sprouts are more erratic in their growth pattern, with some preference for alignment along the long axis of the muscle fiber. The factors that regulate terminal sprout growth are not known.

Age is an important factor in determining sprout growth rate. Aging endplates develop a complex morphology, and longer and more numerous terminal sprouts and nodal sprouts have been found in aged rats compared with young animals.[23] Sprouting in rat soleus, following partial denervation, was significantly reduced in aged animals compared with young adults, but it has also been observed that motor neurons of newborn rats sprout less vigorously. Following partial denervation there is only a twofold increase in the size of neonatal motor units instead of the normal fivefold increase noted in adult muscles.[9] This fivefold increase in motor unit size is independent of the original size and type of the motor unit, and probably represents the limit of the ability of the neuron to supply functional nerve terminals.

Both the rate of nerve terminal degeneration and the maturation of sprouts are different in fast- and slow-twitch muscles. Significant recovery of tension in partially denervated fast-twitch muscles begins 6–7 days after denervation, with nodal sprouting beginning some 2–3 days earlier.[9] In the slow muscle (mouse soleus), relatively little nodal sprouting occurs by the sixth day despite the presence of numerous terminal sprouts. Recovery of tension in slow-twitch muscle takes 9–10 days.

Regeneration of axons into partially denervated muscle may result in double innervation of muscle fibers, with the sprout and the regenerated axon innervating the same fiber.[35] Both histological evidence and tension measurements confirm that both sprouts and regenerated axons innervate the same muscle fibers.[32] The mechanism by which synapses are subsequently eliminated is not well understood but is probably similar to the elimination of multiple innervation of mammalian muscle fibers soon after birth.[2]

Nerve Regeneration. Cajal[12] first showed that viable nerve fibers grow out of the proximal stump of an injured neuron. He showed a latent period between injury and sprouting (about 6 hours) and a further delay (approximately 36 hours) before outgrowth of the axon tips. He showed that growth through the scarred (damaged) area of nerve was slow (approximately 0.25 mm/day) and that invasion of the distal stump of severed nerves may not occur until after 10 or 15 days. Subsequently the rate of regeneration for most fibers was 2–3 mm/day. These observations have been confirmed by many other



B

FIGURE 4. (*continued*)

authors.[55] Recent studies have shown that the earliest sprouts (formed during the first 24 hours) may later degenerate.[31] Following nerve injury in humans, information on the course of regeneration can be provided by serial clinical observations of the location at which Tinel's sign is elicited. Tinel's sign is produced by tapping the skin over a damaged nerve with either a reflex hammer or another blunt object such as a finger. This mechanical stimulation of regenerating sensory axons produces a tingling or shock-like sensation, either locally or in the distribution of the nerve.[64] In detailed studies of human peripheral nerve regeneration, Sunderland[61] found that a severed nerve repaired by simple suture of the cut ends regenerated at a rate of 2.5 mm/day in the upper forearm, 2 mm/day in the lower forearm, and 1 mm/day in the wrist and hand, suggesting a relationship between the distance from the cell body to the lesion and the rate of regeneration. Rates of regeneration are more rapid after a crush injury of nerve (8.5 mm/day in the upper arm, 6 mm/day in the upper forearm, 1–2 mm/day at the wrist, and 1–1.5 mm/day in the hand[61]) than after complete section of a nerve. This difference has been attributed to the lesser degree of distortion of endoneurial tubes in a crush injury. Age affects the rate of nerve regeneration, which is slower in older animals.[5]

Ultrastructurally, the growth cone of regenerating axons is a region of axoplasmic enlargement, rich in smooth endoplasmic reticulum, microtubules, microfilaments, large mitochondria, and lysosomes, from which fine sprouts may extend.[55] Weiss and Edds[65] suggested that elongation of the growth tip may be caused by outward axoplasmic streaming, and the direction of the growth tip may be influenced by mechanical factors in the pathway. Whenever possible, axons regenerate within preexisiting endoneurial tubes, thereby speeding the process of regeneration.

Regeneration of Neuromuscular Junctions. During reinnervation the recovery of neuromuscular junction integrity depends upon the reappearance of active zones to reestablish the conditions for transmitter release. In a study of regenerating frog neuromuscular junctions, nerve terminal length reached 90% of normal by day 16 after nerve crush, but quantal content per unit terminal length recovered to only 67% of normal by 7–12 days, with no subsequent increase.[15] Spontaneous potentials (miniature endplate potentials) recovered more slowly

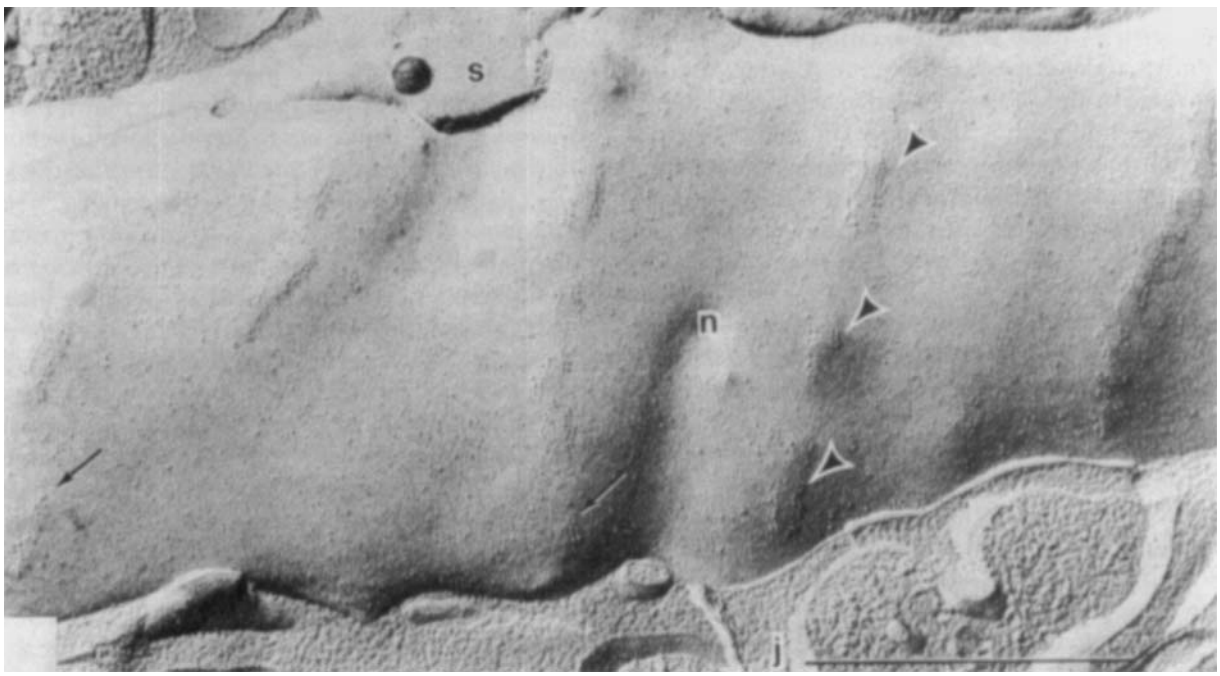


FIGURE 5. Nerve terminal at the secondary stage of regeneration, 28 days after denervation. Some active zone particles form one single or one double row (arrows), with an orientation similar to that of normal. Most active zones, however, are beginning to organize into the normal pattern and orientation at the normal location (arrowheads). There is usually more than one active zone segment aligned with one junctional fold (j), which *is* characteristic of this stage of regeneration. In this muscle EPPs were recorded in every junction studied. Bar = 1 μm. Magnification ×41,000. After Ko.[38]

to only 18% of normal after 30 days, but with normal levels later. The mechanisms underlying the differences in time course of recovery of these different parameters in regenerating junctions have yet to be determined.

Recent freeze-fracture studies of the frog neuromuscular function after nerve crush have demonstrated clusters of large intramembrane particles on nerve terminals opposite junctional folds during early stages of reinnervation (13 days), when nerve stimulation evoked endplate potentials (EPPs) in a few muscle fibers (Fig. 4).[38] When nerve stimulation produced EPPs in every muscle fiber (19 days), a nerve terminal was present in every junctional gutter, and active zones were shorter than normal but lined up in double rows as seen in normal material (Fig. 5). Subsequently, the length of active zones approached normal size after 480 days (Fig. 6).

There is precise localization of active zones across from junctional folds, both in normal tissue and during regeneration. These observations confirm those of Sanes et al.,[54] who showed that regeneration of active zones occurs precisely opposite the basal lamina at junctional folds even after muscle fibers have been removed.

Finally, dimples at an active zone are thought to represent release of one quantum of transmitter.[36] The presence of dimples at clusters of active zone particles early in regeneration suggests that primitive active zones are functional, a fact that is confirmed by the EPPs recorded at the same time.[38] Thus the sequence of regeneration at the neuromuscular junction depends upon initially establishing active zone particles on nerve terminals that gradually mature into the form and function of normal active zones.

Nerve Conduction Velocity in Nerve Regeneration. Following injury to rabbit peroneal nerve, Cragg and Thomas[14] found that conduction velocities proximal to the lesion were reduced by 10% after 25 days and by 20% after 50–100 days. Conduction velocities returned to normal after 150 days and remained so beyond 400 days. One study showed that after successful reinnervation in man, maximal motor conduction velocity proximal to the site of nerve section returns to normal.[3] However, the question of whether slowing of conduction velocity occurs proximal to a compressive nerve lesion in humans remains controversial (see the recent review[29]). In both experimental and human studies of regenerating nerves distal to a lesion, persistent reduction of conduction velocity by 20–50% has

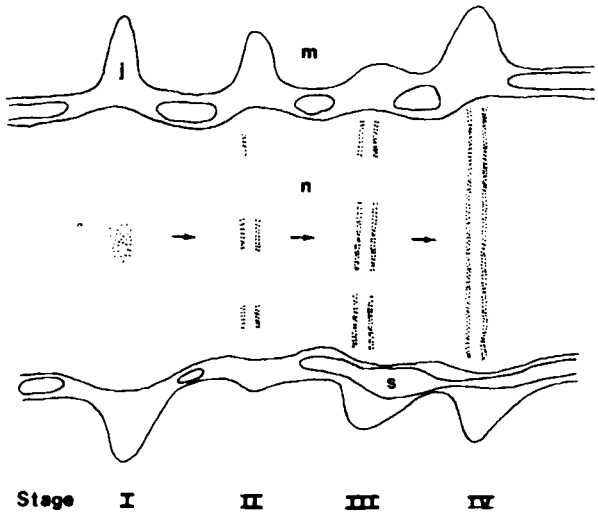


FIGURE 6. A schematic diagram showing the sequence of regeneration of the active zone during stage I (the initial stage), stage II (the secondary stage), stage III (the elongation stage), and stage IV (the late stage). Details in text. Drawings are not to scale. j, junctional fold; m, muscle fiber; n, nerve terminal; s, Schwann cell. After Ko.[38]

been reported. However, in at least one study of experimental animals and in long-term studies of human nerve injuries,[16] motor nerve conduction velocity recovered to greater than 90% of normal after periods in excess of 2 years.

Reinnervation in Human Muscle. Clinical electromyographic techniques have been utilized to examine motor unit potentials during reinnervation following a complete nerve section and subsequent regeneration from the proximal stump of the severed axons. As pointed out by Kugelberg,[39] such motor units have an extremely compact arrangement, with grouped orientation of the muscle fibers and a smaller territory as well as a higher muscle fiber density.[39] During an early stage of regaining voluntary activity, nascent potentials are seen, sometimes including as few as one or two muscle fibers. Motor unit potentials that are seen first during nerve regeneration may resemble fibrillations because they often represent the activity of only a single muscle fiber as the earliest manifestation of a nascent motor unit potential.[6]

After section and suture of human median nerves at the wrist, the earliest responses to motor nerve stimulation were obtained after 3–4 months.[11] At that time muscle action potential amplitudes were 1% of control with prolonged latencies that were 3–6 times normal (Fig. 7). At this time no clinical activity was observed in the muscle (abductor pollicis brevis), but a discrete recruitment pattern was

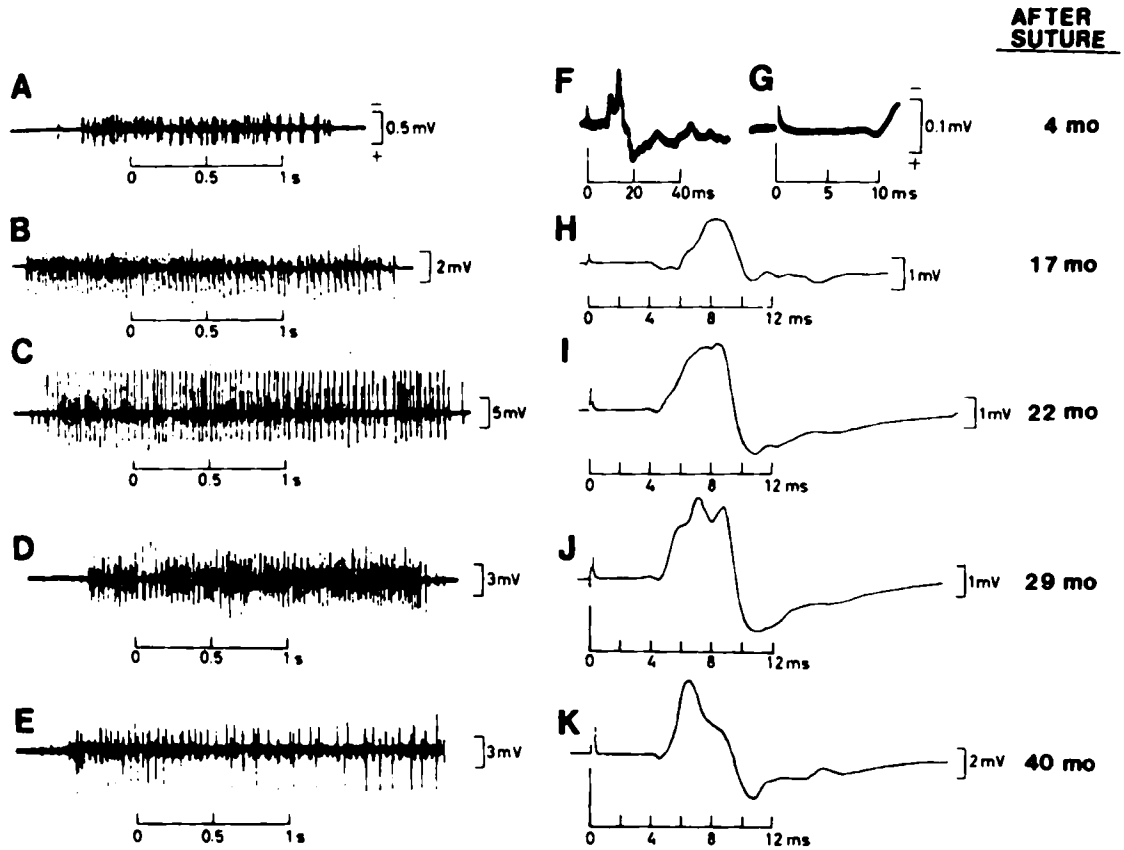


FIGURE 7. Recruitment pattern during full voluntary effort of the abductor pollicis brevis muscle (**A–E**) and action potentials (**F–K**) evoked by maximal stimuli to the nerve at wrist at different times after end-to-end suture of the median nerve. The first response (**F**), evoked 4 months after suture, was recorded with a coaxial needle electrode placed at the site of maximum response; **G** was recorded with a faster sweep speed; **H–K** were recorded with a wire electrode that traversed the cross section of the muscle. The patient was 20 years old. After Buchthal and Kuhl.[11]

observed at electromyography. Motor unit potentials were increased in duration, with increased polyphasia reflecting reinnervation. After 40 months the evoked muscle action potential amplitude reached the lower limit of the normal range, and latency was 50% prolonged, with nearly normal clinical force. At this stage the recruitment pattern was still discrete, and motor unit potential amplitudes were increased up to 300% with durations increased up to 125% compared with the study at 4 months (Fig. 8). Fibrillations were still present in 4–9 sites of the muscle up to 40 months after suture. These data suggest that after regeneration occurs for a limited number of motor units, the subsequent increase in force (after 8–12 months) is attributable to sprouting and enlargement of existing motor units.

Using single-fiber electromyography it is possible to determine the variation in firing of one component of the motor unit with respect to another (jitter). The jitter in normal motor units rarely exceeds 30 μsec, whereas in nascent motor unit potentials the jitter may be of the order of several hundred microseconds, suggesting a precarious neuromuscular transmission safety factor at the newly formed synapse.[19] Blocking (neuromuscular transmission failure) may occur intermittently.

Motor unit potential recruitment is also disturbed in patients recovering from a complete nerve transection. Milner-Brown and his colleagues[47] studied patients recovering from a complete transection of the ulnar nerve 6–24 months after repair. At both 6 and 24 months, the recruitment of motor unit potentials was random with respect to the increased force of voluntary contraction. In keeping with the size principle, smaller motor units were recruited first in the control extremity. During the late phase of regeneration, motor unit potentials, twitch tensions, and overall muscle strength reached normal values. However, motor unit recruitment was still not linear with respect to the correlation between threshold force of voluntary contraction and motor unit twitch tension as found in normal extremities.

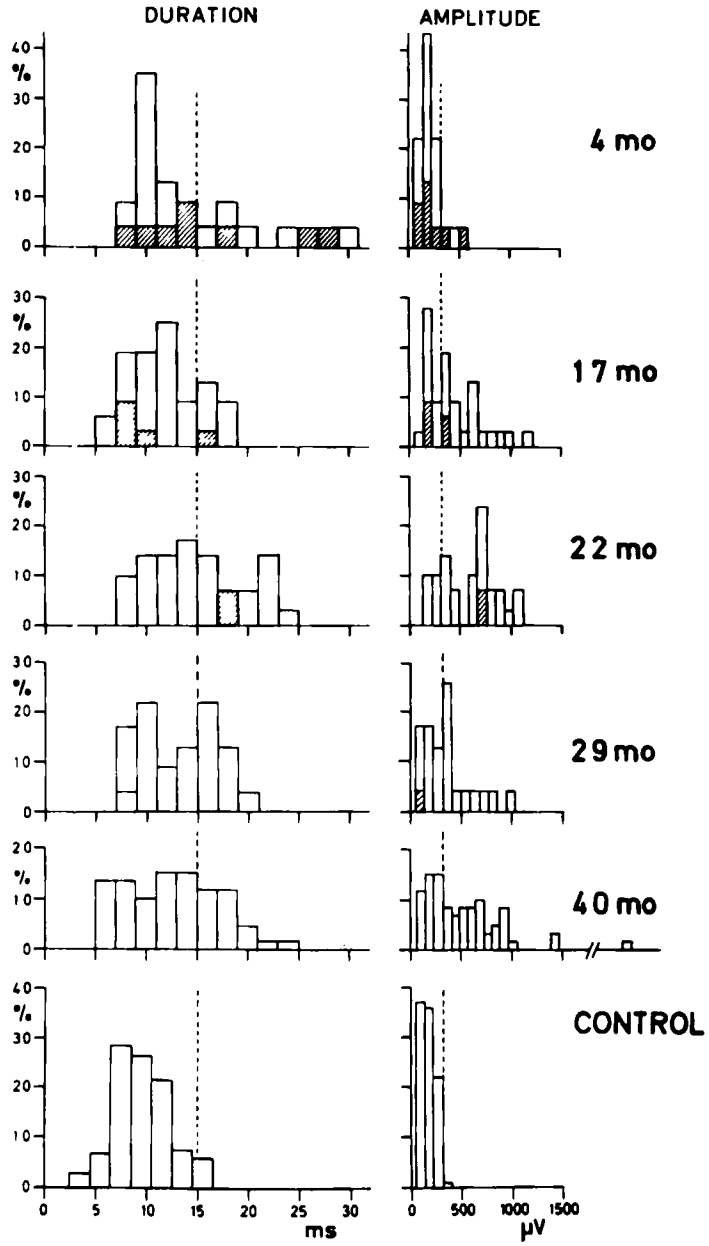


FIGURE 8. Percentage distribution of the duration and the amplitude of motor unit potentials sampled in abductor pollicis brevis muscle during voluntary effort at different times after end-to-end suture of the median nerve at wrist. The striped columns indicate the incidence of polyphasic potentials. Below: 13 control subjects, matched for age. The dashed lines represent the upper 95% confidence limits. The patient was 20 years old. After Buchthal and Kuhl.[11]

CONCLUSION

A complex sequence of events follows injury to a peripheral motor nerve. The pattern of changes at the neuromuscular junction and in the muscle membrane is now reasonably clear, although underlying mechanisms are not. The failure of neuromuscular transmission is a particularly important phenomenon, the pathophysiological basis for which has more completely unfolded in recent years. The pattern of regeneration and the basis for recovering motor function is relatively well worked out. Finally, substantial prognostic information may be obtained after an appropriate time interval through analysis of the evoked compound muscle action potential in response to stimulation of the distal nerve segment. A complete knowledge of these electrophysiological events and their time course is essential both for the clinician as well as the clinical electromyographer.

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